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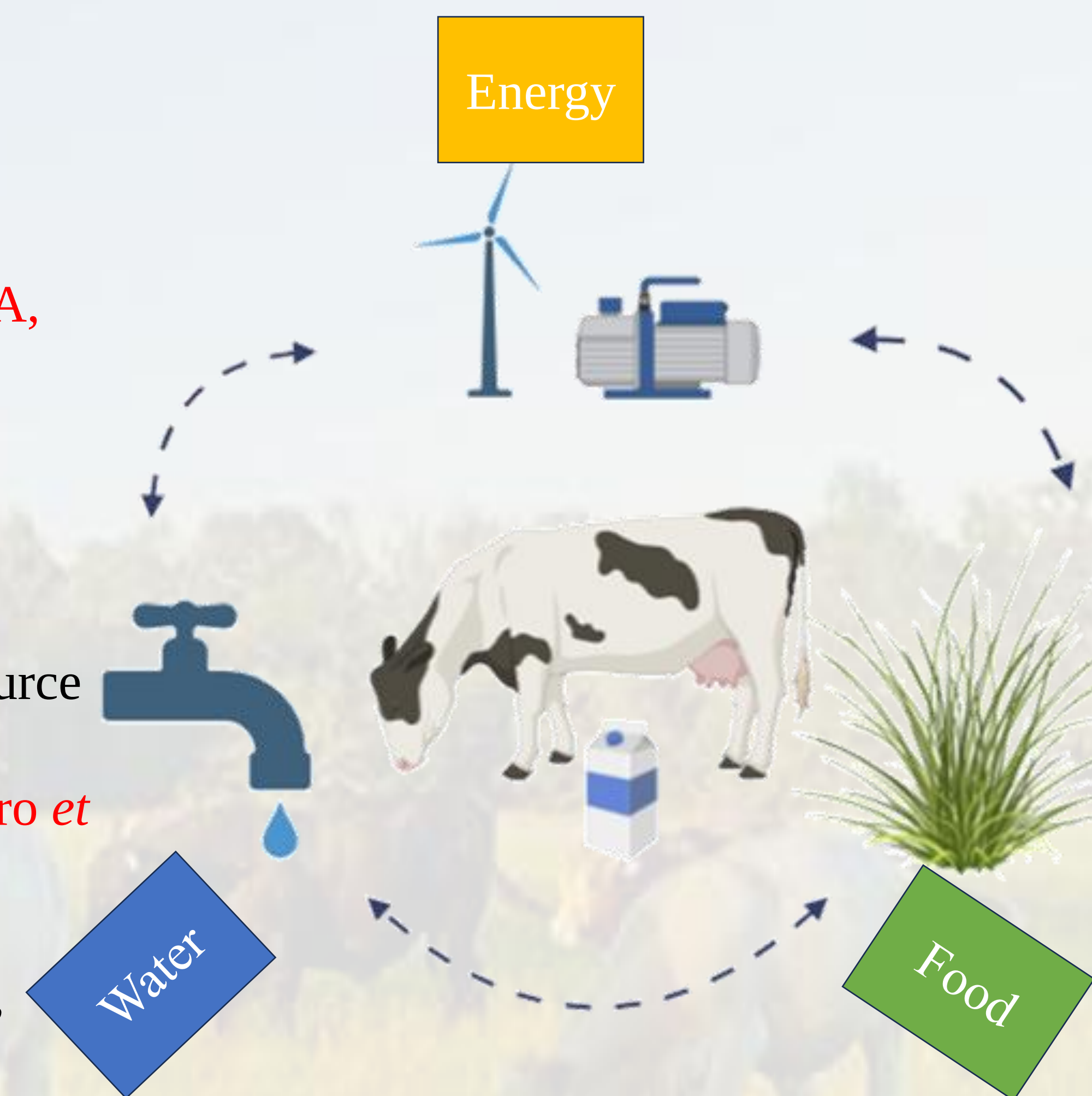
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SUMMARY: Milk production in the U.S. faces mounting challenges from resource limitations and environmental pressures. This study uses system dynamics to model a representative mid-size U.S. dairy farm (1,000 cows), focusing on interactions among water, energy, and feed systems. A causal loop diagram was developed to identify key feedback loops and system behaviors. Findings show strong interdependence between the subsystems, where changes in one impact the others. Future work will involve stock-and-flow modeling and further simulating various scenarios using farm-level and national data, suspected scenario testing will highlight the opportunities to improve efficiency, sustainability, and profitability

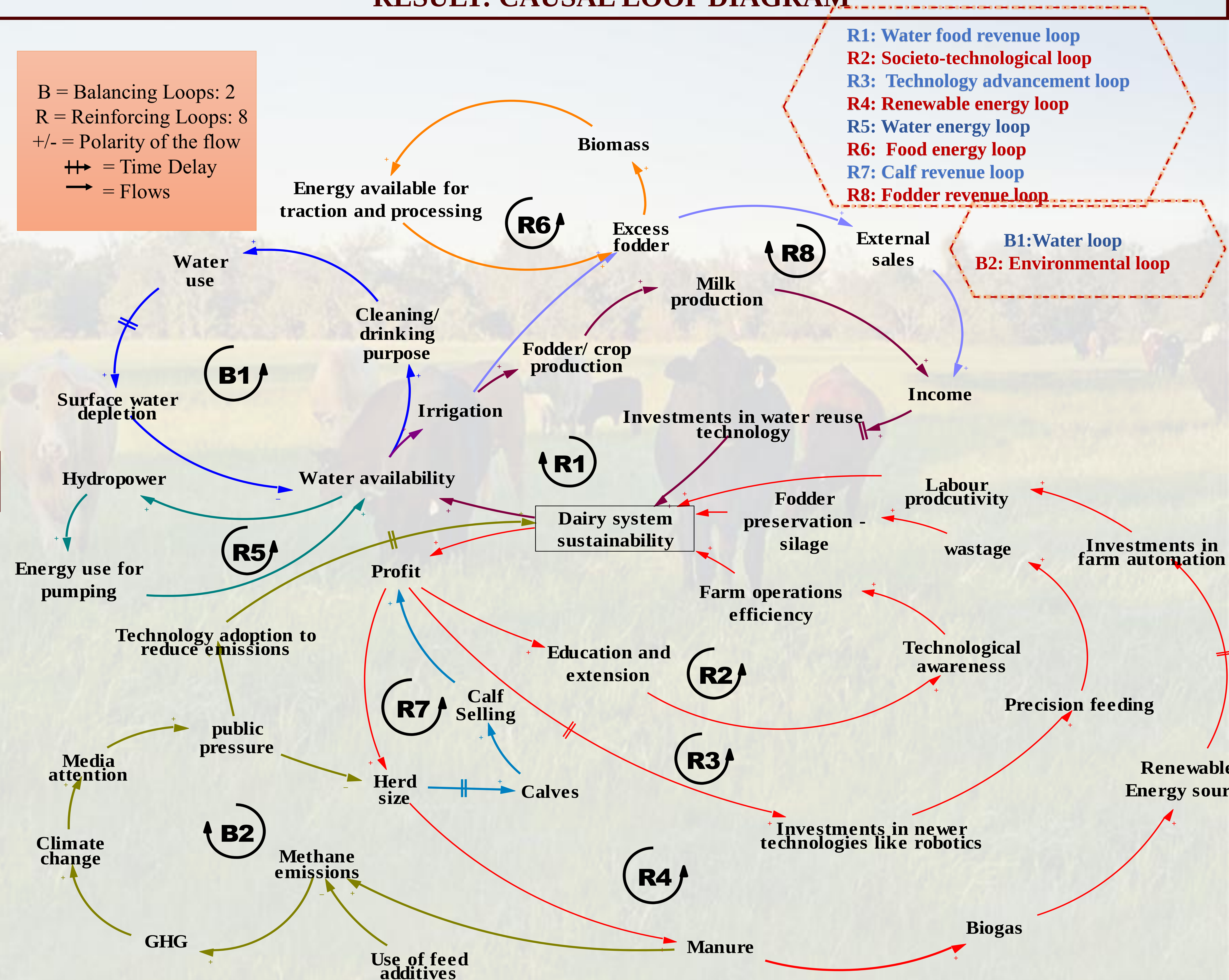
Dynamic hypothesis: Adoption of integrated resource strategies, such as precision irrigation, renewable energy use and nutrient recycling will strengthen positive reinforcing loops among water, energy, and food systems, leading to greater sustainability and resilience of the dairy farm.

RESULT: CAUSAL LOOP DIAGRAM

- US. population: 0.345B people (2024) to 0.347B (2025)- 0.54% annual growth. (Worldometer)
- Dairy cow population: 0.03% increase from last year (USDA, 2025)
- Raising populations- resource scarcity (Bretschger, 2013)
- Need for Water Energy Food Nexus (WEF) : Optimize resource use, Enhance sustainability, Improve resilience, Support decision making, Promote circular systems (Entrena-Barbero et al., 2023)
- System dynamics: behaviour of complex systems over time, simplified versions of reality.

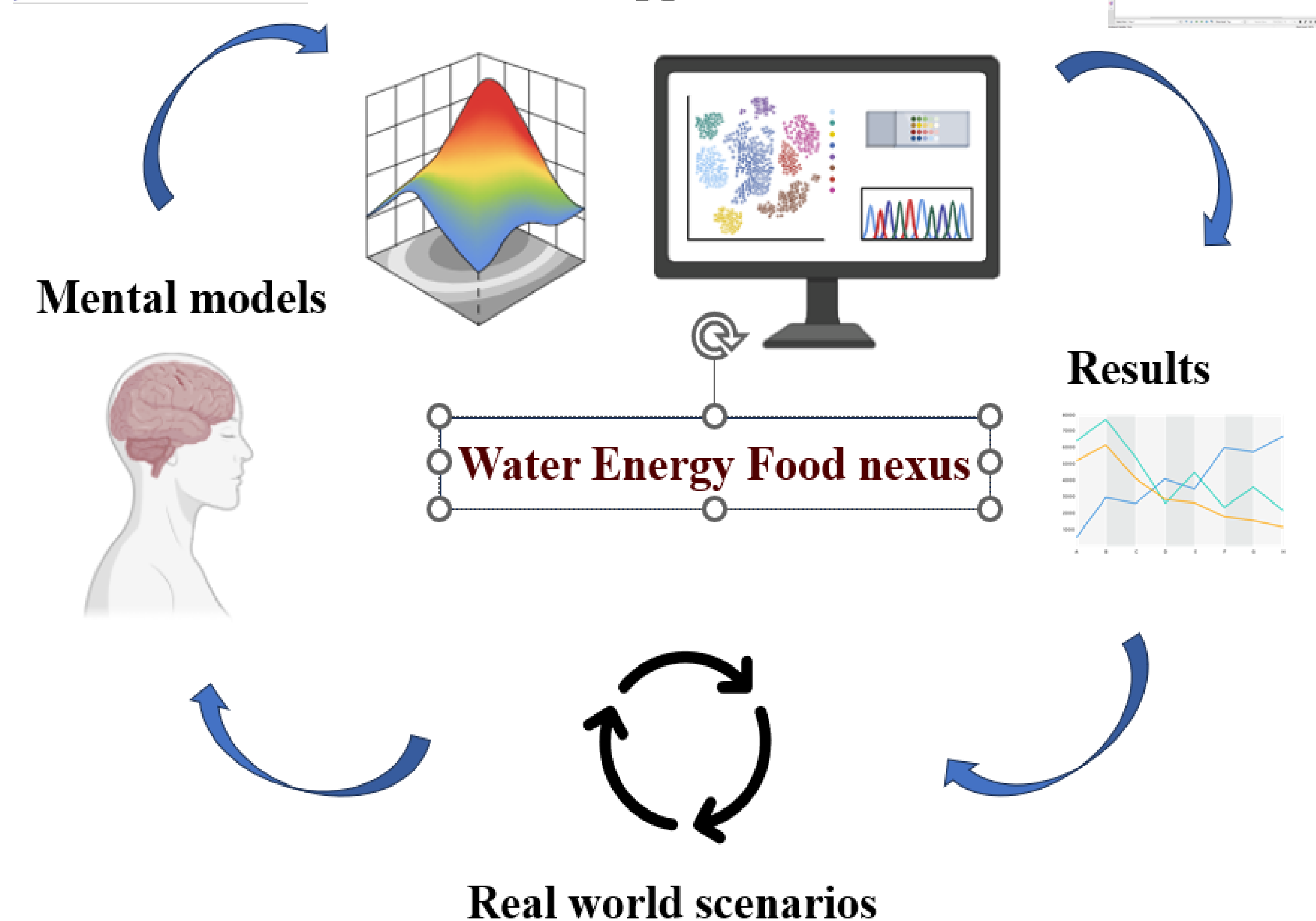


B = Balancing Loops: 2
 R = Reinforcing Loops: 8
 +/- = Polarity of the flow
 ⇨ = Time Delay
 → = Flows



FUTURE IMPLICATIONS

VENSIM Dataset collection, analyzing variables and software approximation



- Stock and flow diagram
- Dynamic model simulation
- Validation of scenarios with real world data
- Transition of this model to agent-based modeling

Scenario	Water Usage	Energy Usage	GHG Emissions
Technological advancements in energy efficiency	No significant change	30% reduction	15% reduction
Improvements in water management	40% reduction	5% increase	5% reduction
Climate adaptation strategies	25% reduction	10% reduction	20% reduction

Table 1: Hypothetical scenario analysis

